

Variable Length

```
In [2]: def sum(a,b):
        c = a + b
        return c
        sum(2,3)
```

Out[2]: 5

```
In [3]: def sum(a,b):
        c = a + b
        return c
        sum(5,6,7,8)    # we define 2 arguments but calling 4 values
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[3], line 4
      2 c = a + b
      3 return c
----> 4 sum(5,6,7,8)

TypeError: sum() takes 2 positional arguments but 4 were given
```

```
In [4]: def sum(a, *b):
        c = a + b
        return c
        sum(5,6,7,8)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[4], line 4
      2 c = a + b
      3 return c
----> 4 sum(5,6,7,8)

Cell In[4], line 2, in sum(a, *b)
      1 def sum(a, *b):
----> 2 c = a + b
      3 return c

TypeError: unsupported operand type(s) for +: 'int' and 'tuple'
```

```
In [5]: def sum(a, *b): # 1st argument is fixed but for 2nd argument
        print(type(a))
        print(type(b))
        sum(5,6,7,8)
```

```
<class 'int'>
<class 'tuple'>
```

```
In [6]: def sum(a, *b): # 1st argument is fixed & we fetch each value from the tuple & we c
        c = a
        for i in b:
```

```

    c = c + i
    print(c)
    sum(5,6,7,8,9,10,11,100)

```

156

kwargs

```

In [7]: def person(name, *data):
        print(name)
        print(data)
        person('Vihari', 3, 'Nandan', 903050)

```

```

Vihari
(3, 'Nandan', 903050)

```

```

In [8]: def person(name,*data):
        print('name')
        print(data)

        person('Vihari', age = 3, home_place = 'southcity', mob =903050)

```

```

-----
TypeError                                Traceback (most recent call last)
Cell In[8], line 5
      2     print('name')
      3     print(data)
----> 5 person('Vihari', age = 3, home_place = 'southcity', mob =903050)

TypeError: person() got an unexpected keyword argument 'age'

```

```

In [9]: def person(name, **data):
        print('name')
        print(data)

        person('Vihari', age = 3, home_place = 'southcity', mob =903050)
        # we got error as keyword argument thats why we add another *

```

```

name
{'age': 3, 'home_place': 'southcity', 'mob': 903050}

```